



Lithium Clinical Guideline

Scope

Site	Department, division or operational area	Applicable to
Royal Perth Bentley Group (RBPB)	Mental Health Division All Clinical Areas	Medical, Nursing, Pharmacy

HIGH RISK MEDICATION

Purpose

The purpose of this guideline is to provide information and guidance on the safe and effective use of lithium.

Key alert

Lithium is a high-risk medication with a narrow therapeutic index. In treating patients with lithium, consideration **must** be given to appropriate patient selection, administration and monitoring requirements to minimise the risk of lithium toxicity often associated with supratherapeutic levels but also possible at therapeutic levels.

Links within document

- [Appendix I: Lithium Monitoring Form](#)

Links to relevant documents/resources

- [RBPB High Risk Medication Policy](#)
- [Therapeutic Guidelines](#)
- [Australian Medicines Handbook](#)
- [AusDI](#)
- [Stahl Online](#)
- [The Maudsley Prescribing Guidelines in Psychiatry](#)

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Version: 2.0

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General information and risks

The decision to treat using lithium should be based on individual patient parameters, including diagnosis, suitability based on medication history and concomitant therapy, comorbidities, and other risk factors such as ability to comply with lithium regime and monitoring requirements.

Lithium has been indicated for the following disorders:

- Treatment for Bipolar Affective Disorder (BPAD)
- Augmentation therapy in refractory unipolar depression
- Management of schizoaffective disorder and as an adjuvant treatment in chronic schizophrenia
- Lithium is more effective in preventing manic/hypomanic/ mixed episodes than depressive episodes ^{1,2,3}.

Contraindications

- **Hypersensitivity** to lithium or any component of its formulation.
- **Severe or significant renal impairment** – lithium is predominantly excreted by the kidneys therefore the risk of lithium toxicity is increased in severe or significant renal impairment associated with medical conditions or concomitant nephrotoxic medications (e.g. diuretics without appropriate dose adjustments). See also [Medication interactions](#) and Lithium toxicity under [Adverse effects](#) sections.
- **Severe sodium depletion** and conditions associated with **hyponatraemia** (e.g. Addison's disease, dehydration, severely debilitated patients and patients on low sodium diets).
- **Severe or significant cardiovascular disease.**

Lithium may cause conduction sinus node dysfunction and conduction blocks resulting in arrhythmias (e.g. sick sinus syndrome) and other electrocardiogram (ECG) changes such as QTc prolongation.

- **Frank hypothyroidism**¹.

Precautions

- **Pregnancy** –TGA Pregnancy Category D lithium may increase the risk of congenital heart defects.
- **Breastfeeding** – lithium enters the breastmilk.
- **Older adults** – use with care in older adults as renal function may be reduced.
- **Surgery** – fasting and fluid/physiological changes can affect serum lithium levels.
- **Electroconvulsive Therapy (ECT)** – lithium may increase risk of delirium. Lithium dose modification required prior to ECT.
- **Serotonin toxicity** – lithium can increase risk of serotonin toxicity when used in combination with other serotonergic medications ^{1,2}.
- **Psoriasis** – may be exacerbated or precipitated by lithium.
- **Obesity** – Lithium clearance is increased in obese patients.

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Medication interactions

Medication interactions are extensive and therefore a comprehensive list is beyond the scope of this document. Some of the major medication interactions that may lead to lithium toxicity are listed below.

Refer to appropriate medication references for further information or a Clinical Pharmacist for advice.

Concomitant use of lithium with any of the following medications **requires closer monitoring of serum lithium concentration, renal function and clinical effects**. Consider use of an alternative agent where possible and adjust the dose of lithium when necessary.

- **Medications that affect renal clearance may increase lithium levels**
 - Angiotensin converting enzyme (ACE) inhibitors and angiotensin II receptor antagonists (may develop within a month to over several weeks, or when switching between agents). Use with caution in the elderly
 - Loop diuretics and thiazide diuretics (may occur within first 10 days)
 - Nonsteroidal anti-inflammatory drugs (NSAIDs) (may occur within a few days to several months). Includes over the counter medications
- **Medications that contain sodium may decrease lithium levels**
 - Urine alkalinisers (e.g. Ural® sachets)
- **Serotonergic medications**
 - In combination with lithium may lead to serotonin toxicity
- **Concurrent antipsychotic medications**
 - Rapid dose increase of lithium and antipsychotics together may increase risk of neurotoxicity (risk varies between antipsychotic agents)^{1,2,3}.

Adverse effects

Lithium toxicity

^{1,2,3}

The typical signs and symptoms associated with lithium toxicity varies dependent on serum lithium levels.

- **Signs and symptoms of mild to moderate toxicity:**
 - Drowsiness, blurred vision, increasing diarrhoea, nausea, vomiting, anorexia, muscle weakness, drowsiness, apathy, ataxia, flu-like illness.
- **Signs and symptoms of severe toxicity:**
 - Increased muscle tone, hyperreflexia, myoclonic jerks, coarse tremor, dysarthria, disorientation, psychosis, seizures, coma, QT-interval prolongation and death.

Escalation of care

Where a patient demonstrates signs and symptoms of toxicity (i.e. mild, moderate or severe), complete the following:

1. Nurse, Pharmacist or Medical Officer (MO) to withhold lithium (noting "W" on *WA Hospital Medication Chart [HMC]*).
2. MO to check the serum lithium level (note time of last dose) and renal function as soon as possible from the time of escalation.
3. MO to clinically assess the patient for risk factors and determine whether lithium doses should continue to be withheld, adjusted or ceased.
4. MO to repeat serum levels as needed.

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Adverse effects – Lithium toxicity cont'd

Signs and symptoms of toxicity are more common at serum levels **above 1.5 mmol/L**, however, may occur at a lower therapeutic level of **1.2 mmol/L** in older adult patients and those with neurological conditions such as epilepsy.

Signs and symptoms of toxicity may indicate possible overdose. **A Toxicologist is to be consulted if poisoning is suspected.**

Lithium can cause **nephrotoxicity**, and long-term irreversible **neurotoxicity**.

Other adverse effects

Table 1: Adverse effects (often dose/level dependent)²

Very common (greater than 10%)	Common (between 1 to 10%)	Infrequent (0.1 to 1%)	Rare (less than 0.1%)
• Tremor • Metallic taste • Nausea • Diarrhoea • Epigastric discomfort • Polydipsia • Polyuria	• Weight gain • Fluid retention • Fatigue • Headache • Vertigo • Acne • Psoriasis • Leucocytosis • Hyperparathyroidism • Nephrotoxicity • Hypothyroidism (usually asymptomatic) • Benign T wave changes on ECG	• Memory impairment • Hair loss	• Renal impairment • Blurred vision • Confusion • Arrhythmias • Hyperthyroidism

Lithium initiation

Patients commenced on lithium must undergo appropriate baseline and ongoing clinical and laboratory investigations aimed at optimising therapy and minimising risk of toxicity.

Pre-initiation work-up

^{1,3}

Prescribers are to obtain personal and family medical history including thyroid function, previous heart disease, renal disease and a best possible medication history.

Nursing/medical staff are to perform baseline investigations as listed in [Table 2](#). Results should be recorded or marked as checked on the *RFBG Lithium Monitoring Form* or in the patient's healthcare record where the form does not allow for recording of the results.

Table 2. Baseline investigations³

Full blood count Thyroid function tests (TFTs)
Urea and electrolytes (U&Es), including calcium, phosphate and magnesium Creatinine
Fasting blood glucose level (BGL) – <i>recorded on BGL Chart</i>
Weight, body mass index (BMI) and waist circumference
ECG
Beta HCG (or other confirmation of pregnancy status) lithium is a human teratogen. *Ensure adequate contraception, and consultation with treating team.

Prescribing and ongoing management

Prescribing of lithium should be individualised to the patient and dependent on indication for use.

Annotate the *WA HMC* to alert staff when routine lithium bloods are due/laboratory tests – (refer to **R PBG Prescribing Policy**).

Dosage and administration

^{1,2,3,4}

- The usual oral starting dose for adults is 500 mg nocte and for patients over 65 years it is 250 mg nocte.
- Dose should be titrated against target serum levels, clinical response along with patient risk factors i.e. weight, comorbidities (e.g. renal impairment) and concomitant medications as appropriate (refer to [Therapeutic medication monitoring](#) section).
- The total daily dose of lithium should not exceed 2500 mg.
- Single night-time dosing with slow release (SR) tablets may improve adherence and allow easier serum lithium level monitoring.
- Twice daily dosing should be spaced by 12 hours (i.e. morning and night).
- It is recommended that lithium be taken with food.
- It may take up to 3 weeks to observe a therapeutic effect following initiation.

Available product presentations

^{1,2,4}

- Lithium **carbonate immediate release (IR)** 250 mg tablet (Lithicarb®).
- Lithium **carbonate slow release (SR)** 450 mg tablet (Quilonum®).
- Lithium **citrate** oral solution BP **IR** 127 mg per 1 mL (Auspman®) Special order product. Approved for patients in the mental health setting at risk of diverting oral tablets.
- Lithium **citrate** per label 127.2mg/mL = lithium carbonate 50mg/mL

Ongoing investigations

Nursing/medical staff are to ensure patients maintained on lithium must undergo regular investigations along with monitoring of serum lithium levels as recommended in Table 3. Results should be recorded or marked as checked on the *RPBG Lithium Monitoring Form* or in the patient's healthcare record where the form does not allow for recording of the results.

Refer to [Therapeutic medication monitoring](#) section for further information on monitoring serum lithium levels.

Table 3: Ongoing investigations (Inpatient & Outpatient)^{2,3}

Recommended investigations	Recommended frequency
Serum lithium levels (referenced against dosage times)	Repeat every 3 to 6 months (on stable dosing)
Full blood count Thyroid function tests (TFTs) - including TSH, T3 and T4	Repeat after 3 months and then every 6 months
U&Es Creatinine	Repeat every 3 to 6 months
BMI, weight and waist circumference	Repeat every 3 to 6 months
Fasting blood sugar levels	Repeat every 6 months
Calcium (including ionised calcium) Parathyroid hormone	Annually
ECG	Annually (<i>dependent on pre-existing cardiac issues or risk</i>)
Beta HCG (or other confirmation of pregnancy status)	Ensure adequate contraception, and consultation with treating team.

The frequency of monitoring should be increased to three monthly if the patient has a co-existing medical condition, such as chronic kidney disease, or if there is a decline in renal function. Additionally, more frequent monitoring is required if the patient is on medications that may interact with lithium.

Therapeutic medication monitoring and dose adjustments

^{1,2,3}

Lithium has a narrow therapeutic index and target serum levels are based on indication for use and patient risk factors. Consider adding a Psychiatric Online Service Information System (PSOLIS/WebPSOLIS) alert (medication alert).

Measuring lithium serum levels

- Levels should be taken **12 hours after the last dose**

For patients on once daily dosing, the 12-hour post-dose level may be **higher** compared to levels where the same daily dose is given as two separate doses (i.e. morning and evening).

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Frequency of monitoring

It is recommended that lithium levels be checked by nursing/medical staff as follows:

- **On admission** to hospital for patients on lithium
- **Following initiation** – 5 to 7 days after commencing treatment, then once weekly for the first four weeks (for inpatients)
- **After each dose change** – check level at 5 to 7 days post dose change
- **While on maintenance** - every 3 months after a stable therapeutic concentration has been achieved
- **More frequent monitoring is recommended when there are:**
 - Changes in
 - fluid intake or signs of illness which may impact fluid balance
 - diet – in particular where there is an increase in sodium intake
 - mental state
 - Concurrent use of interacting medications. Refer to [Medication interactions](#)
 - Signs and symptoms of possible lithium toxicity. Refer to Lithium toxicity in the [Adverse effects](#) section.

Nursing staff are to record observations on the *Lithium Monitoring Form* as directed and notify the prescriber of any abnormal reading. The MO is to:

- Complete timely, ongoing assessment of required monitoring parameters
- Devise a monitoring plan to ensure the patient's ongoing care is comprehensively understood by the entire care team.

Target levels and dose adjustments

Refer also to information on usual therapeutic ranges outlined on the *RPBG Lithium Monitoring Form*.

- **Acute mania**
 - Target serum level: 0.8-1.2 mmol/L.
 - Once remission occurs (7-14 days), gradually decrease to maintenance dose, aim for BPAD prophylaxis lithium levels (below).
- **BPAD prophylaxis**
 - Target serum level: 0.4-1 mmol/L (0.6-0.8 mmol/L if BPAD stabilised).
 - Dosage will vary from one individual to another. Usual maintenance dose is in between 250 mg to 1200 mg daily.
- **Augmentation of antidepressant therapy**
 - Target serum level: 0.6-1 mmol/L.
- **Older adult**
 - Target serum level: 0.4-0.7 mmol/L.
- **Special considerations**
 - Reduce dose and monitor more closely in patients with renal impairment.
 - Where serum creatinine levels are observed to be rising, and decreased creatinine clearance/acute renal impairment is suspected, seek expert advice (contact Renal Physician).
 - The lower end of the therapeutic range is recommended for older adults who are at a greater risk of cognitive impairment and hyponatremia and closer monitoring is recommended.

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Cessation of therapy

During dose reduction and for 3 months after lithium treatment is stopped, monitor the person closely for early signs of mania and depression.

Abrupt cessation of lithium can lead to relapse of mania or depression within a few weeks to months in many patients. Therefore, when ceasing lithium, it should be withdrawn slowly over 1 to 3 months. The cessation of lithium should be done under the supervision of the treating doctor ^{1,3}.

Discharge planning

The patient's treating team are to ensure handing over of important information to future carers to ensure continuity of care. It is the responsibility of the treating teams (inpatient or community) to ensure routine monitoring is completed until care has been formally handed over to a GP/other health service.

Patient education and counselling

Upon commencement of lithium therapy, all patients (including their carers where appropriate), **must** be provided with detailed information and education on lithium treatment and a record of the counselling made in the patient healthcare record (and/or relevant patient electronic health records).

Patients **must** be given written information such as the **WA Department of Health CHOICE Medication leaflet** to support the verbal counselling.

Patient counselling **must** include possible **contributors and symptoms of lithium toxicity** and when to seek **urgent medical review** for possible dose adjustment or temporary cessation of lithium.

Staff should also ensure that patients/carers have achieved a satisfactory level of understanding on the information provided.

Compliance monitoring

Compliance with this Guideline will be monitored via the following:

1. Annual RPBG Mental Health (MH) Division Audit reported by Divisional Quality and Safety meetings.
2. 6 monthly community mental health lithium monitoring audit (for inpatient and outpatient consumers).
3. Datix Clinical Incident Management System (CIMS).
4. Combined Hazard or Incident Reporting (CHoIR) notifications /Notifiable Incident Reporting to the Office of the Chief Psychiatrist, if appropriate.
5. Compliments and Complaints.

Facilitator

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North Metropolitan Health Service Mental Health [Medication: Lithium Prescribing and Management](#)

WA Country Health Service [Specialised Medication- Lithium \(Adult Patients\) Guideline](#)

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Related policies, practice standards, clinical guidelines

Policies and procedures SharePoint

- [High Risk Medication Policy](#)
- [Pre-Admission Anaesthetic Assessment Service \(PAAS\): processes for Patient Medication Review Standard Operational Procedure \(SOP\) \(RPH\)](#)
- [Peri-Operative and Peri-Interventional Medication Management Clinical Guideline](#)
- Prescribing Policy

Department of Health Policy Hub

- [High Risk Medication Policy](#)
- [Mental Health Medication Information Printable Consumer Leaflets](#)

Other

- Mental Health Act 2014
- [Charter of Mental Health Care Principles \(MHA 2014\)](#)

Related national standards

ACSQHC NSQHS Standards 2nd Edition (2021)

Standard 4: Medication Safety

Standard 5: Comprehensive Care

Standard 6: Communicating for Safety

Standard 8: Recognising and Responding to Acute Deterioration

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Appendix I: Lithium Monitoring Form



EMR322170

Please use I.D. label or block print

ROYAL PERTH BENTLEY GROUP MENTAL HEALTH DIVISION LITHIUM MONITORING FORM INITIATION THERAPY (INPATIENT ONLY) WARD: _____ DOCTOR: _____	SURNAME _____ URN _____ GIVEN NAMES _____ D.O.B. _____ SEX _____
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Record individual results on the form. *Where this form does not allow for recording of results document in the patient's healthcare record. Refer to the RPBG Lithium Clinical Guideline for further information.

Lithium Commencement Date: ____ / ____ / ____ Height (for BMI): ____ m

	Baseline	Day 7	Day 14	Day 21	Day 28	3 months	6 months	9 months	12 months
Date									
Lithium dose									
Lithium level									
Time (hours) after dose									
*U&Es / Phosphate / Magnesium									
*FBC									
Fasting BGL									
Serum Creatinine									
Creatinine Clearance									
*TFTs									
*ECG									
*Calcium / PTH (Parathyroid Hormone)									
Weight									
BMI									
Waist Circumference									
Beta HCG									
*Adverse Effects Y / N (document in healthcare record)									
Name (Print)									
Signature									

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LITHIUM MONITORING FORM

MR246E

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ROYAL PERTH BENTLEY GROUP MENTAL HEALTH DIVISION LITHIUM MONITORING FORM INITIATION THERAPY (INPATIENT ONLY)		Please use I.D. label or block print	
WARD: _____ DOCTOR: _____	SURNAME _____	URN _____	GIVEN NAMES _____
DOCTOR: _____	D.O.B. _____	SEX _____	

Starting Lithium Dose: Dosages will vary based on indication, refer to relevant medication references.

Target Lithium Levels:

- Acute Mania: 0.8 to 1.2 mmol/L
- Bipolar Disorder Prophylaxis: 0.4 to 1 mmol/L (stabilised 0.6 to 0.8 mmol/L)
- Augmentation of Antidepressant Therapy: 0.4 to 1 mmol/L
- Elderly: 0.4 to 0.7 mmol/L.

When to Assess Lithium Levels:

- 5 to 7 days after commencing treatment
- Once weekly for first four weeks if initiated as an inpatient
- 5 to 7 days after each dose change
- Every three months after stable therapeutic concentration is achieved.

Blood taken to assess lithium level should be taken 12 hours after the last dose.

WARNING: USE IN PREGNANCY - check prescribing guideline

Common Adverse Effects (not complete list):

- GI upset, nausea, diarrhoea, polydipsia, polyuria, nocturia, dry mouth, fine hand tremor, weight gain, altered taste, decreased libido, hypothyroidism, rash/acne/psoriasis, ECG changes, nephrotoxicity, oedema.
- If creatinine trend is rising even if within normal limits seek Renal Physician advice.**

Withhold lithium dose and measure lithium level if any of the following symptoms of toxicity occur:

- Mild to Moderate: drowsiness, apathy, ataxia, confusion, diarrhoea, vomiting, nausea, blurred vision.
- Severe: dysarthria, increased muscle tone, hyperreflexia, myoclonic jerks, coarse tremor, disorientation, psychosis, QT-interval prolongation, seizures, coma.

Drug interactions with lithium (not a complete list, refer to relevant medication references):

Concomitant use of lithium with any of the following medications requires close monitoring of serum lithium concentrations, renal function and clinical effects for extended periods. Medications that:

- INCREASE in lithium levels and risk of toxicity:
 - NSAIDs (including COX-2 inhibitors) - low dose aspirin may be safe
 - ACE inhibitors
 - Angiotensin II Receptor Inhibitors
 - Diuretics
- DECREASE in lithium levels:
 - Sodium Containing drugs (e.g. Ural Sachets), sodium containing foods
- POTENTIAL Side Effects of other medications:
 - Antipsychotics - can cause neurotoxicity with rapid dose increases - monitor closely
- PROLONGATION of effects of the following agents:
 - Neuromuscular Blocking Agents: consider ceasing lithium 2 to 3 days before ECT or other procedures
- SEROTONIN SYNDROME risk when used with the following agents:
 - Serotonergic agents.

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